

Improper integrals – 3

Improper integrals depending on a parameter

Homework

12 сентября 2026 года

Theoretical minimum. Uniform convergence of $\int_a^\infty f(x, t)dx$ means that the tails are small simultaneously for all t . It is checked by the Cauchy criterion, by the Weierstrass majorant, or — when there is no integrable majorant — by the parametric forms of the Dirichlet and Abel tests, which is the only way to handle $\int_0^\infty e^{-tx} \frac{\sin x}{x} dx$ at $t = 0$. Uniform convergence gives continuity of the integral in the parameter, the right to integrate in the parameter under the sign, and, together with uniform convergence of the differentiated integral, the right to differentiate under the sign. Where it fails, everything fails: $\int_0^\infty t e^{-tx} dx$ jumps from 1 to 0 at $t = 0$. For integrals over a finite interval the same statements are in Definite integrals – 8; here everything has to survive an infinite tail or a singular endpoint.

1. Prove that $\int_0^\infty \frac{1 - \cos(tx)}{x^2} dx = \frac{\pi|t|}{2}$ by differentiating in t , and note the subtlety: the differentiated integral $\int_0^\infty \frac{\sin(tx)}{x} dx$ converges only conditionally, so the Weierstrass test is unavailable and the parametric Dirichlet test has to be used.
2. Prove that $J(t) = \int_0^\infty \frac{dx}{(1+x^2)^t}$ is defined and infinitely differentiable for $t > \frac{1}{2}$, compute $J'(1)$, and find the asymptotics of $J(t)$ as $t \rightarrow \frac{1}{2}^+$.
3. Prove that $\int_0^\infty \frac{x^{s-1}}{1+x} dx$ converges exactly for $0 < s < 1$, that the convergence is uniform on $[\delta, 1 - \delta]$, and that the function so defined is continuous there.
4. Prove that $\int_0^\infty \frac{\sin(tx)}{x} dx$ equals $\frac{\pi}{2} \operatorname{sign} t$, that the convergence is uniform on $\{|t| \geq \delta\}$ and not uniform on any neighbourhood of 0, and explain the discontinuity at $t = 0$ in terms of Theorem 2 of the class sheet.

5. Prove the analogue of Dini's theorem: if $f(x, t) \geq 0$ is continuous, $F(t) = \int_a^\infty f(x, t) dx$ converges for every t and is continuous on $[c, d]$, then the convergence is uniform on $[c, d]$.
6. Let $G(t) = \int_0^\infty \frac{\sin(tx)}{x(1+x^2)} dx$ for $t > 0$. Prove that G is twice differentiable, that $G'' - G = -\frac{\pi}{2}$, and find G in closed form; state where uniform convergence is used and where it fails.
7. Prove that Γ is logarithmically convex on $(0, \infty)$ and deduce the Bohr–Mollerup characterisation: a positive function on $(0, \infty)$ with $f(1) = 1$, $f(s+1) = sf(s)$ and $\ln f$ convex is Γ .
8. Let f be absolutely integrable on $[0, \infty)$. Prove that $F(t) = \int_0^\infty f(x) \cos(tx) dx$ is continuous on $\mathbb{R}$ and tends to 0 as $|t| \rightarrow \infty$, and show that without absolute integrability both conclusions may fail.
9. Prove the classical identity $(\int_0^t e^{-x^2} dx)^2 + \int_0^1 \frac{e^{-t^2(1+x^2)}}{1+x^2} dx = \frac{\pi}{4}$ by differentiating in t , and deduce $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$. Compare with Integrals – 3, where the same value came out of a differential equation in the parameter.
10. Let $F(t) = \int_0^\infty \frac{\arctan(tx)}{x(1+x^2)} dx$. Prove that F is differentiable for $t > 0$, compute F' , and find $F(t)$ in closed form; state precisely where uniform convergence was used.